

Ordinary Differential Equations

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1 Introduction

The subject of differential equations can be broken into two sub subjects, ordinary differential equations (ODEs) and partial differential equations (PDEs). This lesson will focus on ordinary differential equations and will leave partial differential equations to a future lecture and notes. In that vein lets recall a the derivative $f' : \mathbb{R} \rightarrow \mathbb{R}$ models the change of the function $f : \mathbb{R} \rightarrow \mathbb{R}$. Building on this there is a similar relationship between higher order derivatives.

This implies that an equation involving derivatives of various orders is going to model the change in that mathematical system. (At this point I am speaking about the equations themselves, not yet the physical systems we see in engineering modeled by these equations.)

2 Definitions

Ordinary Derivative

Ordinary Derivative: Let $A \subseteq \mathbb{R}$ and $x_0 \in A$. Suppose $f : A \rightarrow \mathbb{R}$. Then we define the ordinary derivative or just derivative $f' : A \rightarrow \mathbb{R}$ as

$$\frac{d}{dx}f(x) = f'(x) := \lim_{x \rightarrow x_0} \frac{f(x+h) - f(x)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Ordinary Deferential Equation 1

Define an Ordinary Differential Equation: An equation involving one or more ordinary derivatives.

This definition formalizes in a slightly confusing manor. So before formalizing the definition we will define implicit and explicit forms of equations which differ from homogeneous and nonhomogeneous forms of an equation.

Explicit Equations

Define Explicit Equations: Consider a relation between two variables x, y . Say y is dependent on x and x is independent. Then $y = f(x)$. We define an equation of this form to be an explicit equation.

An example of an explicit equation is $y = x^2 + 3x - 2$

Implicit Equations

Define Implicit Equations: Consider a relation between two variables x, y . If the relation can not be solved or is not solved for x in terms of y or visa versa we call this an implicit equations and is of the form $f(x, y) = 0$

An example of an implicit equation is $x^2 + y^2 - \pi = 0$

Ordinary Differential Equation 2

Define an Ordinary Differential Equation: Let V be the vector space with basis $\{x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}\}$ and $F : V \rightarrow V$ and $G : V \rightarrow V$. Then F, G are a functions of x, y and the derivatives of y . In explicit form we have;

$$F(x, y, \frac{dy}{dx}, \dots, \frac{d^{n-1}y}{dx^{n-1}}) = \frac{d^n y}{dx^n}$$

And in implicit form;

$$G(x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}) = 0$$

(Note these represent the same differential equation solved for different variables. Additionally, if the vector space or basis aspects of this definition are unfamiliar or uncomfortable see the linear algebra lectures and notes.)

3 Classifications of ODEs

It is important to be able to classify a differential equation as that tells us what method to use when solving. ODEs have four main classifications of an ODE, linearity, homogeneity, order, and coefficient type.

Linearity is determining weather an ODE is linear or nonlinear. If the function F as defined in the definition of an ODE is linear over y and its derivatives then the ODE is linear. If however, F is nonlinear over y or any of its derivatives, then the ODE is nonlinear. (If the ODE is nonlinear, your best bet is to go straight to numerical approximation. There ARE ODEs that are nonlinear and have known analytic solutions, but these are rare and it is generally not worth your time to attempt to find such a solution. Unless explicitly asked by an assignment as outside the educational setting the likely hood of your specific ODE having an analytical solution is low.)

Examples of linearity:	A	$y'' + 3y' - y = 0$	Linear
	B	$y' + \ln(y) = x$	Nonlinear
	C	$\pi y'' + y = x^3$	Linear
	D	$\sum_{i=0}^n a_i(x)y^{(i)} = f(x)$	Linear
	E	$g(x, y, y', \dots, y^n) = f(x)$	Nonlinear $\forall g$ such that g is nonlinear

(for more information concerning linearity see the Wikipedia page)

Homogeneity is determined off the component of F concerning x . Suppose F is written implicitly as $F(x, y, y', \dots, y^n) = g(x, y, y', \dots, y^n) + f(x)$ where g has no term solely dependent on x . Then if $f(x) = 0$ the ODE is homogeneous. If $f(x) \neq 0$ then the ODE is nonhomogeneous.

Examples of homogeneity:	A	$y'' + 3y' - y = 0$	Homogeneous
	B	$y' + \ln(y) = x$	Nonhomogeneous
	C	$\pi y'' + xy = 0$	Homogeneous

The order of a ODE is the highest order derivative of the ODE. Again looking at our general implicit definition $F(x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}) = 0$ the order would be n .

Examples of order:	A	$y'' + 3y' - y = 0$	second order
	B	$y' + \ln(y) = x$	first order
	C	$\pi y'' + xy = 0$	second order

The coefficient type refers to the terms involving y and its derivatives. This set of terms is $\{a_i(x)y^i | 0 \leq i \leq n\}$. The classification is constant or non constant coefficient. Let $A = \{a_j \in V | j \neq i\}$. An ODE is constant coefficient $\iff a_i(A) = c_i$ where $c_i \in \mathbb{R}, \forall 0 \leq i \leq n$. Therefore if any $a_i(A)$ is non constant in x (meaning it has some function of x) then the ODE is non constant coefficient.

Examples of coefficient type:	A	$y'' + 3y' - y = 0$	constant coefficient
	B	$y' \ln(y) = x$	constant coefficient
	C	$\pi y'' + xy = 0$	non constant coefficient
	D	$xyy' + y = 4$	non constant coefficient

4 Methods of solving an ODE

There are a number of various technics used to solve ODEs in which this document will only mention by name. For more depth on each of the techniques feel free to google the techniques named, or consult the notes in this series on on those techniques. The main techniques are:

Separation of variables
Variation of parameters
Characteristic roots

Undetermined coefficients

Ingratiating factor method

Laplace transforms

5 Engineering Applications

There are many applications of ODEs ranging from PID controllers to thermal applications. Anything that can be modeled as a changing system can be modeled with differential equations. We care about this subject then as it allows us to build numerous models for a variety of systems.